
Library Inventory System

Team 9

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CS157A Intro to Database Management Systems

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March 17, 2026

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Functional Requirements

Below is a list of the Library Inventory System's core features, broken down by who has access to which feature.

1. Borrow and Librarian

- a. Create account (username, password)
 - i. Users should also be able to give their first name, last name, phone number, address, age, and so on to create their very own library account.
- b. Book search (potential filters - genre)
 - i. Admin should be able to view a database of all the current books in the system
 - ii. To search for books, buyers/borrowers should be able to search a book by title, publication and potentially even by adding publication date filters.
- c. Login
 - i. In addition to this, the buyer should also be able to login to their existing library account

2. Borrower View

- a. View book description
 - i. After searching for a book, borrowers may select a book to view its description
 - ii. Descriptions include title, author, publication date, condition, availability, and synopsis
- b. Borrow book

-
- i. Upon selecting a book, if it is available, the borrower may choose to borrow the book
 - ii. Borrowed books will have a given return date before late fees are placed, which can be extended by renewing
 - c. Return book
 - i. Borrowers may return any books they have borrowed
 - ii. Returned books will be removed from the borrower's account once an admin has checked the book in
 - d. Renew book
 - i. Borrowers should be able to renew a book they have currently borrowed
 - ii. These books could be renewed unless they are not already on hold for someone else
3. Library view (Admin related functionalities):
- a. Adding new books, Books: title, author
 - i. Administrators should be able to take the book title, author, publication, and other details and seamlessly add it to the system so that other users can borrow it.
 - ii. After reloading the page, these changes should persist in the view.
 - b. Deleting a book entirely from system
 - i. Admin should be able to delete a book from the library such that it will no longer be available to borrow.
 - ii. Changes should persist after committing in the admin view.
 - c. Add hold

-
- i. Administrators may add a hold to a book that a borrower has requested to borrow
 - ii. The book will be exclusively available to the borrower for a set period of time before it goes to the next borrower
 - iii. Borrowers will be notified of the book's hold duration and pickup location
 - d. Set borrow time
 - i. Upon creating an administrator account, the user will be able to input the length of time, in days, books are able to be borrowed. If they choose not to manually input a time, the default borrow time will be 21 days.
 - e. Renew book/hold
 - i. The administrator can renew a borrower's time with a book.
 - ii. The borrower's time will be reset to the borrow time set by the library admin, or default if no borrow time was set.
 - f. Check in book after its return
 - i. When the borrower returns a book, the system will flag that the book is available again for the borrower.
 - g. Lend book
 - i. This functionality allows the librarian to mark when books have been borrowed. This makes them unavailable to borrow, but a hold can be added.

Entities

Library

Each library can be uniquely identified by a Library_ID, which serves as the primary identifier for each library in the system. This ensures that every library branch or location can be distinguished from others within the database. The Name attribute stores the name of the library, allowing users and administrators to easily recognize the specific library branch. The Address attribute records the physical location of the library so that users can identify where the library is situated.

Additionally, each library includes a Phone Number, which provides a way for users or administrators to contact the library for inquiries or support. Together, these attributes describe essential information about each library entity in the system. By storing the library's unique identifier, name, location, and contact number, the database can efficiently manage and reference different libraries within the overall library management system.

Users (split into borrower and admin)

Each user in the system is uniquely identified by a User_ID, which serves as the primary key. The First_Name and Last_Name store the user's name, while Address and Phone number contact details. The Username and Password are used for login and authentication. These attributes allow users to create accounts, log in, and interact with the library system. Users protect their account with a password and also have details such as their first name, last name, address, and phone number.

Admin

Each administrator in the system is identified by a User_ID, which also serves as a foreign key referencing the Users entity. The EmployeeID uniquely identifies the administrator within the

library system, while the Permissions attribute defines the actions the administrator is allowed to perform. These attributes enable administrators to manage system operations such as adding books, handling holds, and configuring settings. The Admin entity is a subtype of Users in an ISA relationship, meaning every administrator is also a user with additional privileges.

Borrower

Each borrower in the system is identified by a User_ID, which also serves as a foreign key referencing the Users entity. The CardNum uniquely identifies the borrower's library card. The maxBorrowed attribute defines the maximum number of books a borrower can have at one time, while numBorrowed tracks the current number of books borrowed. These attributes help manage borrowing limits and track user activity within the system. The Borrower entity is a subtype of Users in an ISA relationship, meaning every borrower is also a user with borrowing-specific attributes.

Book

Each book in the system is uniquely identified by a Book_ID, which serves as the primary key for the book record in the database. The Title attribute stores the name of the book so that users can easily identify and search for it. Author attribute records the person or people who wrote the book. Publisher stores the name of the publishing company responsible for publishing the book. The Genre attribute categorizes the book into a specific literary or subject category, which helps organize the library collection. Synopsis attribute provides a short summary or description of the book's content. Finally, the Publication_Date attribute records the date when the book was officially published. All these different features could help in the filtering and searching component of the database.

Inventory

Each physical copy of a book in the system is uniquely identified by a Copy_ID, which serves as the primary key for the inventory record. The Book_ID attribute acts as a foreign key that links each copy to its corresponding book in the Book_Info entity, ensuring that every physical copy is associated with a valid book in the system. The Status attribute indicates the current availability of the book copy, such as whether it is available, borrowed, or on hold.

Loans

Each loan in the system is uniquely identified by a Loan_ID, which serves as the primary key. The User_ID and Copy_ID attributes link (acts as a foreign key) the loan to the borrower and the specific book copy, respectively. The Borrow_Date records when the book was issued, while the Due_Date indicates when it must be returned. The Date_Returned captures when the book is actually returned, and the Renewal_Count tracks how many times the loan has been renewed. These attributes collectively support borrowing, returning, and renewal operations within the system.

Holds

Each hold in the system is uniquely identified by a Hold_ID, which serves as the primary key. The User_ID and Book_ID link the hold to the borrower and the requested book. The Request_Date records when the hold was placed, while the Expiration_Date defines how long the hold remains valid. The Pickup_Location specifies where the book can be collected. These attributes enable the system to manage reservations efficiently.

Settings

Each settings record in the system is uniquely identified by a Setting_ID, which serves as the primary key. The Default_Borrow_Val defines the standard number of days a book can be borrowed, while the Max_Renewals specifies the maximum number of times a loan can be renewed. These attributes allow administrators to control borrowing policies within the system.

Relationships

Contains

Has is the relationship between Inventory (weak) and Library (supporting). Each inventory can only be a part of one library, whereas a library can have multiple book inventories, making it one to many. It uses Inventory's Copy_ID and Library's Library_ID .

Registers

Registers is the relationship between User and Library. A User can only register for one library location, whereas a library can have many users registered, making it one to many. It uses User's User_ID and Library's Library_ID.

Borrows

Borrows is the relationship between Borrower and Loan. A Loan can be borrowed by only one Borrower, whereas a Borrower can borrow many Loans, making it one to many. It uses Borrower's CardNum and Loan's Loan_ID.

LoansOut

Loans out is the relationship between Loan and Inventory. A Loan can only be loaned out by one Inventory, whereas an Inventory can loan out many Loans, making it one to many. It uses Loan's Loan_ID and Inventory's Copy_ID.

Requests

Requests is the relationship between Borrower and Hold. A Borrower can request many Holds, whereas a Hold can only be requested by one Borrower, making it many to one. It uses Borrower's CardNum and Hold's Hold_ID.

Reserves

Reserves is the relationship between Hold and Book. A Hold can only be reserved for one Book, whereas a Book can have many Hold reservations, making it one to many. It uses Hold's Hold_ID and Book's Book_ID.

Approves

Approves is the relationship between Hold and Admin. A Hold can only be approved by one Admin, whereas an Admin can approve many Holds, making it one to many. It uses Hold's Hold_ID and Admin's Employee_ID.

Modifies

Modifies is the relationship between Settings and Admin. An Admin can modify multiple Settings, making it one to many. It uses Setting's Setting_ID and Admin's Employee_ID.

Has

Contains is the relationship between Book and Inventory. A Book can have many copies in the Inventory, whereas an Inventory can have copies of one Book, making it many to one. It uses Book's Book_ID and Inventory's Copy_ID.

Edits

Contains the Relationship between Admin and Inventory. Edits has the attribute Type which contains what editing takes place. It uses Copy_ID from Inventory and EmployeeID and User_ID from Admin.

Schema

Entities

Library (Library_ID (PK), Name, Address, Phone_Number)

Users (User_ID (PK), Username, Password, First_Name, Last_Name, Phone, Address)

Borrower (CardNum, User_ID (PK/FK), maxBorrowed, numBorrowed)

Admin (EmployeeID, User_ID (PK/FK), Permissions)

Book (Book_ID (PK), Title, Author, Publisher, Publication_Date, Genre, Synopsis)

Inventory (Copy_ID (PK), Status, Library_ID (FK))

Loans (Loan_ID (PK), Borrow_Date, Due_Date, Date_Returned, Renewal_Count)

Holds (Hold_ID (PK), Request_Date, Expiration_Date, Pickup_Location)

Settings (Setting_ID (PK), Max_Renewals, Default_Borrow_Val)

Relationships

Has(Book_ID, Copy_ID)

Reserves(Hold_ID, Book_ID)

Registers(User_ID, Library_ID)

Borrows(CardNum, User_ID, Loan_ID)

LoansOut(Loan_ID, Copy_ID)

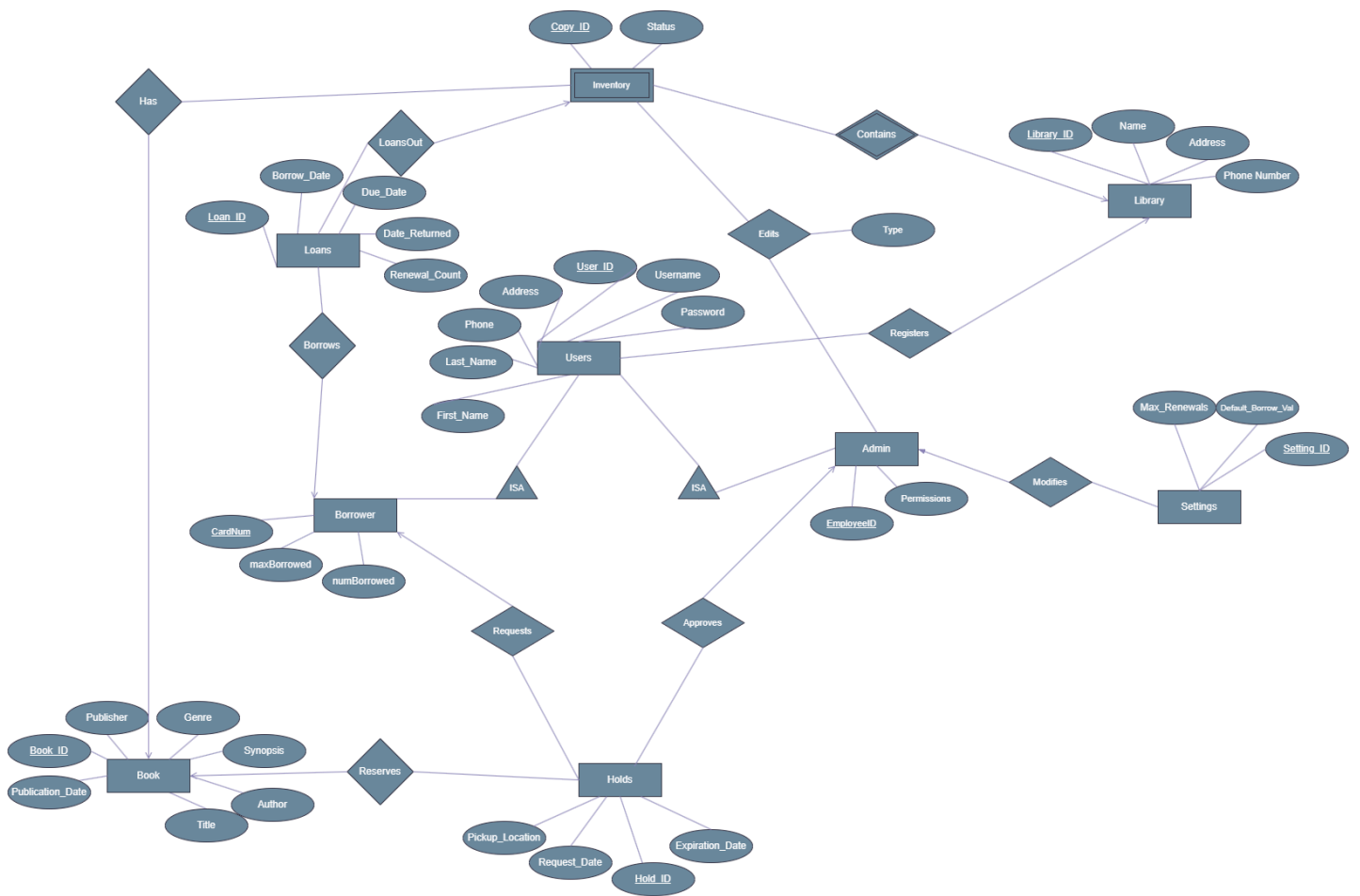
Requests(CardNum, User_ID, Hold_ID)

Approves(Hold_ID, Employee_ID, User_ID)

Admin_Modifies(Setting_ID, Employee_ID, User_ID)

Edits (Copy_ID, Employee_ID, User_ID, Type)

ERD



SQL Workbench

The screenshot shows the MySQL Workbench interface with the 'Admin' table selected in the Schemas pane. The table schema is displayed in the Object Info pane, showing columns: User_ID (int PK), EmployeeID (int), and Permissions (varchar(255)). The main query editor contains the query: `SELECT * FROM Team9LibSys.Admin;`. The Result Grid shows 10 rows of data, with columns: User_ID, EmployeeID, and Permissions. The Action Output pane shows the execution of the query, indicating 10 rows affected and 0 duplicates.

User_ID	EmployeeID	Permissions
11	5001	All
12	5002	Read/Write
13	5003	Read Only
14	5004	All
15	5005	Read/Write
16	5006	All
17	5007	Read/Write
18	5008	Read Only
19	5009	All
20	5010	Read/Write

The screenshot shows the MySQL Workbench interface with the 'Admin_Modifies' table selected in the Schemas pane. The table schema is displayed in the Object Info pane, showing columns: Setting_ID (int PK), User_ID (int), and EmployeeID (int). The main query editor contains the query: `SELECT * FROM Team9LibSys.Admin_Modifies;`. The Result Grid shows 10 rows of data, with columns: Setting_ID, User_ID, and EmployeeID. The Action Output pane shows the execution of the query, indicating 10 rows returned and 0.00041 sec / 0.0000... sec.

Setting_ID	User_ID	Employee_ID
1	11	5001
2	12	5002
3	13	5003
4	14	5004
5	15	5005
6	16	5006
7	17	5007
8	18	5008
9	19	5009
10	20	5010

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Context... Snippets

SCHEMAS

Filter objects

- Tables
 - Admin
 - Admin_Modifies
 - Approves
 - Book
 - Borrower
 - Borrows
 - Edits
 - Has
 - Holds
 - Inventory
 - Library
 - Loans
 - LoansOut
 - Registers
 - Requests
 - Reserves
 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 SELECT * FROM Team9LibSys.Approves;

100% 1:1

Result Grid

Hold_ID	User_ID	Employee_ID
1	11	5001
2	12	5002
3	13	5003
4	14	5004
5	15	5005
6	16	5006
7	17	5007
8	18	5008
9	19	5009
10	20	5010

Approves 1

Action Output

Time	Action	Response	Duration / Fetch Time
✓ 42 10:50:32	SELECT * FROM Team9LibSys.Approves LIMIT 0, 1000	10 row(s) returned	0.00041 sec / 0.0000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Book Context... Snippets

SCHEMAS

Filter objects

- Tables
 - Admin
 - Admin_Modifies
 - Approves
 - Book
 - Borrower
 - Borrows
 - Edits
 - Has
 - Holds
 - Inventory
 - Library
 - Loans
 - LoansOut
 - Registers
 - Requests
 - Reserves
 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 SELECT * FROM Team9LibSys.Book;

100% 1:1

Result Grid

Book_ID	Title	Author	Publisher	Publication_Da...	Genre	Synopsis
1	The Hobbit	J.R.R. Tolkien	Allen & Unwin	1937-09-21	Fantasy	A hobbit goes on an adventure.
2	1984	George Orwell	Secker & Warburg	1949-06-08	Dystopian	Big Brother is watching.
3	Dune	Frank Herbert	Chilton Books	1965-08-01	Sci-Fi	Spice must flow.
4	Foundation	Isaac Asimov	Gnome Press	1951-05-01	Sci-Fi	Psychohistory saves the galaxy.
5	Neuromancer	William Gibson	Ace	1984-07-01	Cyberpunk	Hacking the matrix.
6	Catch-22	Joseph Heller	Simon & Schuster	1961-11-10	Satire	War is absurd.
7	Beloved	Toni Morrison	Knopf	1987-09-02	Historical	A haunting legacy.
8	The Color Purple	Alice Walker	Harcourt	1982-01-01	Fiction	A story of resilience.
9	Fahrenheit 451	Ray Bradbury	Ballantine	1953-10-19	Dystopian	Burning books.
10	Brave New World	Aldous Huxley	Chatto & Windus	1932-01-01	Sci-Fi	A soma-fueled future.

Book 1

Action Output

Time	Action	Response	Duration / Fetch Time
✓ 43 10:50:36	SELECT * FROM Team9LibSys.Book LIMIT 0, 1000	10 row(s) returned	0.00050 sec / 0.0000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Book Borrower

SCHEMAS

Filter objects

- Tables
 - Admin
 - Admin_Modifies
 - Approves
 - Book
 - Borrower
 - Borrows
 - Edits
 - Has
 - Holds
 - Inventory
 - Library
 - Loans
 - LoansOut
 - Registers
 - Requests
 - Reserves
 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 SELECT * FROM Team9LibSys.Borrower;

Result Grid

User_ID	CardNum	maxBorrowed	numBorrowed
1	1001	5	1
2	1002	5	2
3	1003	5	0
4	1004	5	3
5	1005	5	1
6	1006	5	0
7	1007	5	4
8	1008	5	1
9	1009	5	0
10	1010	5	2
HULL	HULL	HULL	HULL

Borrower 1

Action Output

Time	Action	Response	Duration / Fetch Time
✓ 44 10:50:40	SELECT * FROM Team9LibSys.Borrower LIMIT 0, 1000	10 row(s) returned	0.00043 sec / 0.000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Book Borrower Borrows

SCHEMAS

Filter objects

- Tables
 - Admin
 - Admin_Modifies
 - Approves
 - Book
 - Borrower
 - Borrows
 - Edits
 - Has
 - Holds
 - Inventory
 - Library
 - Loans
 - LoansOut
 - Registers
 - Requests
 - Reserves
 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 SELECT * FROM Team9LibSys.Borrows;

Result Grid

User_ID	Loan_ID	CardNum
1	1	1001
2	2	1002
3	3	1003
4	4	1004
5	5	1005
6	6	1006
7	7	1007
8	8	1008
9	9	1009
10	10	1010
HULL	HULL	HULL

Borrows 1

Action Output

Time	Action	Response	Duration / Fetch Time
✓ 45 10:50:43	SELECT * FROM Team9LibSys.Borrows LIMIT 0, 1000	10 row(s) returned	0.00037 sec / 0.0000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Book Borrower Borrowers Edits Context... Snippets

SCHEMAS

Filter objects

- Tables
 - Admin
 - Admin_Modifies
 - Approves
 - Book
 - Borrower
 - Borrowers
 - Edits
 - Has
 - Holds
 - Inventory
 - Library
 - Loans
 - LoansOut
 - Registers
 - Requests
 - Reserves
 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 SELECT * FROM Team9LibSys.Edits;

Limit to 1000 rows

Result Grid

Copy_ID	User_ID	Employee_ID	Type
1	11	5001	Status Update
2	12	5002	Condition Log
3	13	5003	Location Change
4	14	5004	Status Update
5	15	5005	Condition Log
6	16	5006	Location Change
7	17	5007	Status Update
8	18	5008	Condition Log
9	19	5009	Location Change
10	20	5010	Status Update
HULL	HULL	HULL	HULL

Edits 1

Action Output

Time	Action	Response	Duration / Fetch Time
46 10:50:47	SELECT * FROM Team9LibSys.Edits LIMIT 0, 1000	10 row(s) returned	0.00060 sec / 0.000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Book Borrower Borrowers Edits Has Context... Snippets

SCHEMAS

Filter objects

- Tables
 - Admin
 - Admin_Modifies
 - Approves
 - Book
 - Borrower
 - Borrowers
 - Edits
 - Has
 - Holds
 - Inventory
 - Library
 - Loans
 - LoansOut
 - Registers
 - Requests
 - Reserves
 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 SELECT * FROM Team9LibSys.Has;

Limit to 1000 rows

Result Grid

Book_ID	Copy_ID
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
HULL	HULL

Has 1

Action Output

Time	Action	Response	Duration / Fetch Time
47 10:50:56	SELECT * FROM Team9LibSys.Has LIMIT 0, 1000	10 row(s) returned	0.00045 sec / 0.000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Book Borrower Borrowers Edits Has Context... Snippets

SCHEMAS

Filter objects

- Tables
 - Admin
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 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 SELECT * FROM Team9LibSys.Holds;

Limit to 1000 rows

100% 1:1

Result Grid

Hold_ID	Request_Date	Expiration_Date	Pickup_Location
1	2023-10-11	2023-10-18	Central Library
2	2023-10-12	2023-10-19	North Branch
3	2023-10-13	2023-10-20	South Branch
4	2023-10-14	2023-10-21	East Branch
5	2023-10-15	2023-10-22	West Branch
6	2023-10-16	2023-10-23	Downtown Library
7	2023-10-17	2023-10-24	Uptown Library
8	2023-10-18	2023-10-25	University Branch
9	2023-10-19	2023-10-26	Community Center
10	2023-10-20	2023-10-27	Historic Branch

Holds 1

Action Output

Time	Action	Response	Duration / Fetch Time
48 10:51:03	SELECT * FROM Team9LibSys.Holds LIMIT 0, 1000	10 row(s) returned	0.00040 sec / 0.000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Book Borrower Borrowers Edits Has Context... Snippets

SCHEMAS

Filter objects

- Tables
 - Admin
 - Admin_Modifies
 - Approves
 - Book
 - Borrower
 - Borrowers
 - Edits
 - Has
 - Holds
 - Inventory
 - Library
 - Loans
 - LoansOut
 - Registers
 - Requests
 - Reserves
 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 SELECT * FROM Team9LibSys.Inventory;

Limit to 1000 rows

100% 1:1

Result Grid

Copy_ID	Status	Library_ID
1	Available	1
2	Checked Out	2
3	Lost	3
4	Available	4
5	In Transit	5
6	Available	6
7	Checked Out	7
8	Available	8
9	Damaged	9
10	Available	10

Inventory 1

Action Output

Time	Action	Response	Duration / Fetch Time
49 10:51:06	SELECT * FROM Team9LibSys.Inventory LIMIT 0, 1000	10 row(s) returned	0.0012 sec / 0.00001...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Book Borrower Borrowers Edits Has Context... Snippets

SCHEMAS

Filter objects

- Tables
 - Admin
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 - Borrower
 - Borrowers
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 - LoansOut
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 - Reserves
 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 • SELECT * FROM Team9LibSys.Library;

Limit to 1000 rows

100% 1:1

Result Grid

Library_ID	Name	Address	Phone_Number
1	Central Library	100 Main St	555-0101
2	North Branch	200 North Ave	555-0102
3	South Branch	300 South Blvd	555-0103
4	East Branch	400 East Rd	555-0104
5	West Branch	500 West Ln	555-0105
6	Downtown Library	600 City Center	555-0106
7	Uptown Library	700 High St	555-0107
8	University Branch	800 College Way	555-0108
9	Community Center	900 Park Dr	555-0109
10	Historic Branch	1000 Old Town	555-0110

Library 1

Action Output

Time	Action	Response	Duration / Fetch Time
✓ 50 10:51:09	SELECT * FROM Team9LibSys.Library LIMIT 0, 1000	10 row(s) returned	0.00051 sec / 0.0000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Book Borrower Borrowers Edits Has Context... Snippets

SCHEMAS

Filter objects

- Tables
 - Admin
 - Admin_Modifies
 - Approves
 - Book
 - Borrower
 - Borrowers
 - Edits
 - Has
 - Holds
 - Inventory
 - Library
 - Loans
 - LoansOut
 - Registers
 - Requests
 - Reserves
 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 • SELECT * FROM Team9LibSys.Loans;

Limit to 1000 rows

100% 1:1

Result Grid

Loan_ID	Borrow_Date	Due_Date	Date_Returned	Renewal_Count
1	2023-10-01	2023-10-15	2023-10-14	0
2	2023-10-02	2023-10-16	NULL	1
3	2023-10-03	2023-10-17	2023-10-18	0
4	2023-10-04	2023-10-18	NULL	0
5	2023-10-05	2023-10-19	2023-10-15	0
6	2023-10-06	2023-10-20	NULL	2
7	2023-10-07	2023-10-21	2023-10-20	0
8	2023-10-08	2023-10-22	NULL	0
9	2023-10-09	2023-10-23	2023-10-21	1
10	2023-10-10	2023-10-24	NULL	0

Loans 1

Action Output

Time	Action	Response	Duration / Fetch Time
✓ 51 10:51:12	SELECT * FROM Team9LibSys.Loans LIMIT 0, 1000	10 row(s) returned	0.00044 sec / 0.000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Book Borrower Borrowers Edits Has Context... Snippets

SCHEMAS

Filter objects

- Tables
 - Admin
 - Admin_Modifies
 - Approves
 - Book
 - Borrower
 - Borrowers
 - Edits
 - Has
 - Holds
 - Inventory
 - Library
 - Loans
 - LoansOut
 - Registers
 - Requests
 - Reserves
 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 SELECT * FROM Team9LibSys.LoansOut;

100% 1:1

Result Grid

Loan_ID	Copy_ID
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
HULL	HULL

LoansOut 1

Action Output

Time	Action	Response	Duration / Fetch Time
52 10:51:15	SELECT * FROM Team9LibSys.LoansOut LIMIT 0, 1000	10 row(s) returned	0.00036 sec / 0.000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Book Borrower Borrowers Edits Has Context... Snippets

SCHEMAS

Filter objects

- Tables
 - Admin
 - Admin_Modifies
 - Approves
 - Book
 - Borrower
 - Borrowers
 - Edits
 - Has
 - Holds
 - Inventory
 - Library
 - Loans
 - LoansOut
 - Registers
 - Requests
 - Reserves
 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 SELECT * FROM Team9LibSys.Registers;

100% 1:1

Result Grid

User_ID	Library_ID
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
HULL	HULL

Registers 1

Action Output

Time	Action	Response	Duration / Fetch Time
53 10:51:18	SELECT * FROM Team9LibSys.Registers LIMIT 0, 1000	10 row(s) returned	0.00046 sec / 0.000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Book Borrower Borrowers Edits Has Context... Snippets

SCHEMAS

Filter objects

- Tables
 - Admin
 - Admin_Modifies
 - Approves
 - Book
 - Borrower
 - Borrowers
 - Edits
 - Has
 - Holds
 - Inventory
 - Library
 - Loans
 - LoansOut
 - Registers
 - Requests
 - Reserves
 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 SELECT * FROM Team9LibSys.Requests;

100% 1:1

Result Grid

User_ID	Hold_ID	CardNum
1	1	1001
2	2	1002
3	3	1003
4	4	1004
5	5	1005
6	6	1006
7	7	1007
8	8	1008
9	9	1009
10	10	1010
HULL	HULL	HULL

Filter Rows: Search Edit: Export/Import: Result Grid Form Editor Field Types Query Stats

Requests 1

Action Output

Time	Action	Response	Duration / Fetch Time
54 10:51:21	SELECT * FROM Team9LibSys.Requests LIMIT 0, 1000	10 row(s) returned	0.00049 sec / 0.000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Book Borrower Borrowers Edits Has Context... Snippets

SCHEMAS

Filter objects

- Tables
 - Admin
 - Admin_Modifies
 - Approves
 - Book
 - Borrower
 - Borrowers
 - Edits
 - Has
 - Holds
 - Inventory
 - Library
 - Loans
 - LoansOut
 - Registers
 - Requests
 - Reserves
 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 SELECT * FROM Team9LibSys.Reserves;

100% 1:1

Result Grid

Hold_ID	Book_ID
10	1
9	2
8	3
7	4
6	5
5	6
4	7
3	8
2	9
1	10
HULL	HULL

Filter Rows: Search Edit: Export/Import: Result Grid Form Editor Field Types Query Stats

Reserves 1

Action Output

Time	Action	Response	Duration / Fetch Time
55 10:51:24	SELECT * FROM Team9LibSys.Reserves LIMIT 0, 1000	10 row(s) returned	0.00042 sec / 0.0000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Book Borrower Borrowers Edits Has Context... Snippets

SCHEMAS

Filter objects

- Tables
 - Admin
 - Admin_Modifies
 - Approves
 - Book
 - Borrower
 - Borrowers
 - Edits
 - Has
 - Holds
 - Inventory
 - Library
 - Loans
 - LoansOut
 - Registers
 - Requests
 - Reserves
 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 SELECT * FROM Team9LibSys.Settings;

100% 1:1

Result Grid

Setting_ID	Max_Renewals	Default_Borrow_Val
1	2	14
2	3	21
3	1	7
4	2	14
5	2	14
6	4	28
7	2	14
8	3	21
9	1	7
10	5	30
HALL	HALL	HALL

Settings 1

Action Output

Time	Action	Response	Duration / Fetch Time
✓ 56 10:51:26	SELECT * FROM Team9LibSys.Settings LIMIT 0, 1000	10 row(s) returned	0.00052 sec / 0.0000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQLWorkbench File Edit View Query Database Server Tools Scripting Help

CS157a

Administration Schemas Team9LibSys - Schema Query 2 Admin Admin_Modifies Approves Book Borrower Borrowers Edits Has Context... Snippets

SCHEMAS

Filter objects

- Tables
 - Admin
 - Admin_Modifies
 - Approves
 - Book
 - Borrower
 - Borrowers
 - Edits
 - Has
 - Holds
 - Inventory
 - Library
 - Loans
 - LoansOut
 - Registers
 - Requests
 - Reserves
 - Settings
 - Users
 - Views

Object Info Session

Table: Approves

Columns:

- Hold_ID int PK
- User_ID int PK
- Employee_ID int

1 SELECT * FROM Team9LibSys.Users;

100% 33:1

Result Grid

User_ID	Username	Password	First_Name	Last_Name	Phone	Address
1	jsmith	pass123	John	Smith	555-1001	11 Apple St
2	jdoe	pass123	Jane	Doe	555-1002	12 Berry Ln
3	bobw	pass123	Bob	Williams	555-1003	13 Cherry Ct
4	alice1	pass123	Alice	Johnson	555-1004	14 Data Ave
5	charleb	pass123	Charlie	Brown	555-1005	15 Elm St
6	evad	pass123	Eva	Davis	555-1006	16 Fig Rd
7	frankm	pass123	Frank	Miller	555-1007	17 Grape Dr
8	gracew	pass123	Grace	Wilson	555-1008	18 Hazel Blvd
9	henryt	pass123	Henry	Taylor	555-1009	19 Ivy Ln
10	ivy	pass123	Ivy	Anderson	555-1010	20 Joy St
11	admin1	adminpass	Admin	One	555-2001	Staff Box 1
12	admin2	adminpass	Admin	Two	555-2002	Staff Box 2
13	admin3	adminpass	Admin	Three	555-2003	Staff Box 3
14	admin4	adminpass	Admin	Four	555-2004	Staff Box 4
15	admin5	adminpass	Admin	Five	555-2005	Staff Box 5
16	admin6	adminpass	Admin	Six	555-2006	Staff Box 6
17	admin7	adminpass	Admin	Seven	555-2007	Staff Box 7
18	admin8	adminpass	Admin	Eight	555-2008	Staff Box 8
19	admin9	adminpass	Admin	Nine	555-2009	Staff Box 9
20	admin10	adminpass	Admin	Ten	555-2010	Staff Box 10
HALL	HALL	HALL	HALL	HALL	HALL	HALL

Users 1

Action Output

Time	Action	Response	Duration / Fetch Time
✓ 57 10:51:29	SELECT * FROM Team9LibSys.Users LIMIT 0, 1000	20 row(s) returned	0.00043 sec / 0.000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

References

Project_Proposal and Requirement.pdf. *CS 157A: Introduction to Database Management Systems*, Spring 2026, *San Jose State University*, sjsu.instructure.com.

Project Proposal & Functional Requirements Document- Example.pdf. *CS 157A: Introduction to Database Management Systems*, Spring 2026, *San Jose State University*, sjsu.instructure.com.